Lakshya JEE 2.0 2025

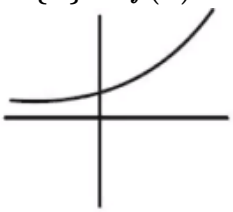
Maths

Relations and Functions

DPP: 9

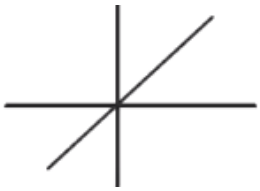
Q1 $[x]+\{x\}=f(x)$. Then graph of $f(x)+1$ is:

(A)

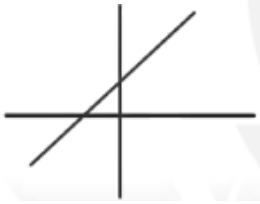


(B)

(C)

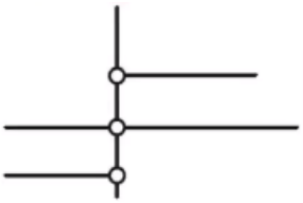


(D)

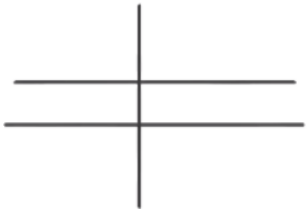


Q2 Which of the following is graph of $\mathrm{Sgn}(e^x)$?

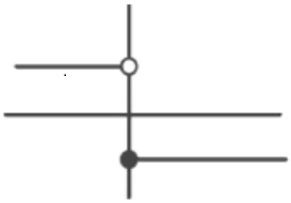
(A)



(B)



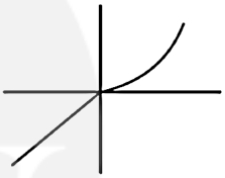
(C)



(D)

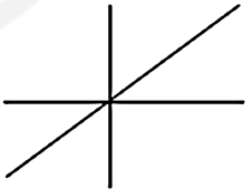
Q3 Draw graph of $\max\{e^x, x\}$

(A)

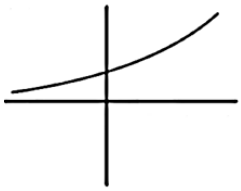


(B)

(C)



(D)



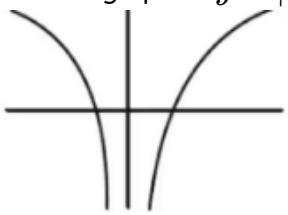
Q4 No. of solutions of equation $\cos x=\frac{x}{10}$ is/are

(A) 6 (B) 7

(C) 8 (D) 9

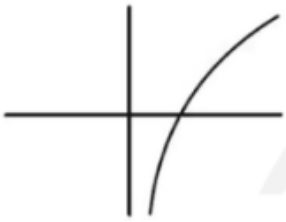
Q5 Which is the graph of $y = |\log|x||$?

(A)

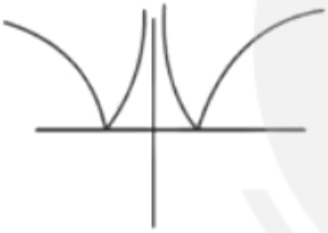


(B)

(C)

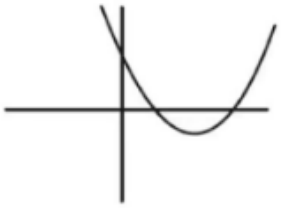


(D)

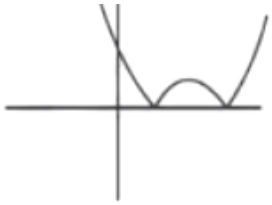


Q6 Which of the following is graph of $|y| = (x-1)(x-2)$?

(A)

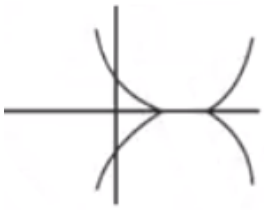


(B)



(C)

(D)



Q7 Graph of $y = |\log(-x)|$ lies in

(A) First quadrant

(B) Second quadrant

(C) Third quadrant

(D) Fourth quadrant

Q8 Which of the following functions is exactly same as $y = |\sin x|$?

(A) $y = \sin|x|$

(B) $y = |\cos x|$

(C) $y = |\sin|x||$

(D) $y = \sin(-x)$

Q9 What are the number of solutions of the equation? $5^{-x} = -|\sin x|$

(A) 0 (B) 1

(C) 2 (D) 3

Q10 Find number of solutions of the following equation $e^x - 3 + 7x = 0$

(A) 0 (B) 1

(C) 2 (D) 3

Q11 If the graph of the function $y = 2^x$ is reflected over the x-axis, the equation of reflected curve is

(A) $y = 2^{-x}$

(B) $y = x^2$

(C) $y = -2^x$

(D) $y = -2^{-x}$

Q12 Function $y = f(x)$ has been shifted 3 units to right and 5 units down. Which one of the following equation represents these changes?
(A) $y = f(x+3) - 5$
(B) $y = f(x-3) + 5$
(C) $y = f(x+3) + 5$
(D) $y = f(x-3) - 5$

Q13 If $||x-1|-5| = 2$, then the number of distinct values of x is
(A) 1 (B) 2
(C) 3 (D) 4

Q14 Draw graph of $f(x) = \ln|x|$

Q15 Draw graph of $f(x) = ||x|-3|$

Q16 Draw graph of $f(x) = 2 - \sqrt{9-x^2}$

Q17 Draw graph of $f(x) = -\{-x\}$

Q18 If $f(x) = |5-|x||$ and $g(x) = 7^{-|x|}$ then number of solution of equation $f(x) = g(x)$ is_____.

Q19 Number of point of intersection of $f(x) = [x]$ and $g\left(x\right) = \frac{x+1}{3}$

Q20 Find number of solution of following equation: $\cos x = |x|$

Q21 Draw graph of $f\left(x\right) = \sqrt{|x|-3}$

Q22 Draw graph of $f(x) = \ln(2-|x|)$

Answer Key

Q1 (D)

Q2 (B)

Q3 (D)

Q4 (B)

Q5 (D)

Q6 (D)

Q7 (B)

Q8 (C)

Q9 (A)

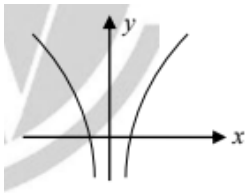
Q10 (B)

Q11 (C)

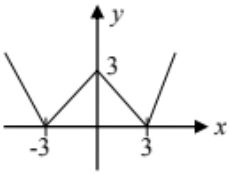
Q12 (D)

Q13 (D)

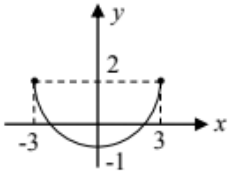
Q14



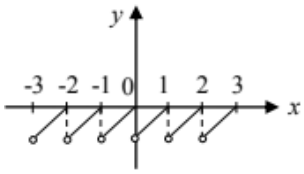
Q15



Q16



Q17

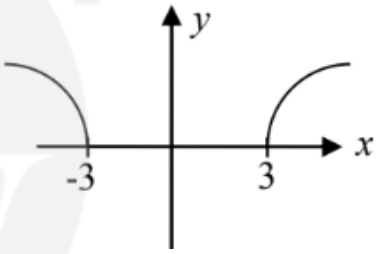
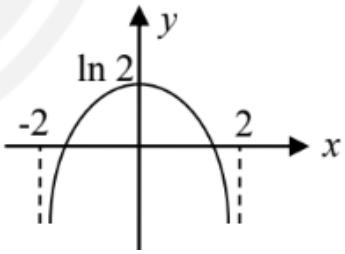


Q18 4 Solution

Q19 0

Q20 2

Q21



Q22